Problem Set 4

1

You are working on a new feature for the smart toilet, which uses electrodes in the seat. This feature can help a user to learn how well their diet is working for them by measuring their body composition (weight, fat, muscle, hydration level, etc.). As a side benefit, the electrodes can be energized to remind the user they have been sitting for too long.

a

Using the simplified model for the user’s body mass and skin shown, derive the equivalent circuit model consisting of 2 resistors and one capacitor. The simple model has the skin-electrode interface modeled as a resistor and the body mass as a capacitor. Assume the skin has no capacitance. Model the cones as resistors and the cylinder as a capacitor.

NOTE: the contact point at the end of the cone has a 1 cm radius.
NOTE: You can use $\sigma_{SKIN} = 0.1\ S/m, \epsilon_{BODY} = 300$

Answer

$$R = \int_0^h \frac{1}{\sigma\pi(\frac{R-r}{h}x+r)^2}\, dx$$

$$R = 636.619772368\ \Omega$$

$$C = \frac{\epsilon A}{L} = 0.167\ nF$$

2

Charge $q_1 = 6\ \mu C$ is located at (1cm, 1cm, 0) and charge q_2 is located at (0, 0, 4cm). What should q_2 be so that $\vec{E}$ at (0, 2cm, 0) has no y component?

Answer

$$q_1(\frac{\vec{R}-R_1}{|\vec{R}-\vec{R}_i|^3}) + q_2(\frac{\vec{R}-R_2}{|\vec{R}-\vec{R}_2|^3})$$

$$q_2 = -\frac{q_1 0.01}{\sqrt{0.0002}^3}\frac{\sqrt{0.002}^3}{0.02}$$

$$q_2 = -94.86\ \mu C$$

3

A horizontal strip lying in the x-y plane is of width d in the y direction and infinitely long in the x direction. If the strip is in air and has a uniform charge distribution ρ_s, use Coulomb's law to obtain an explicit expression for the electric field at a point P located a distance h above the centerline of the strip. Extend your result to the special case where d is infinite and compare it with Eq. (4.25).

✓ **Answer**

We know this will be symmetric in all directions except for upwards, so we only need to integrate the z component.

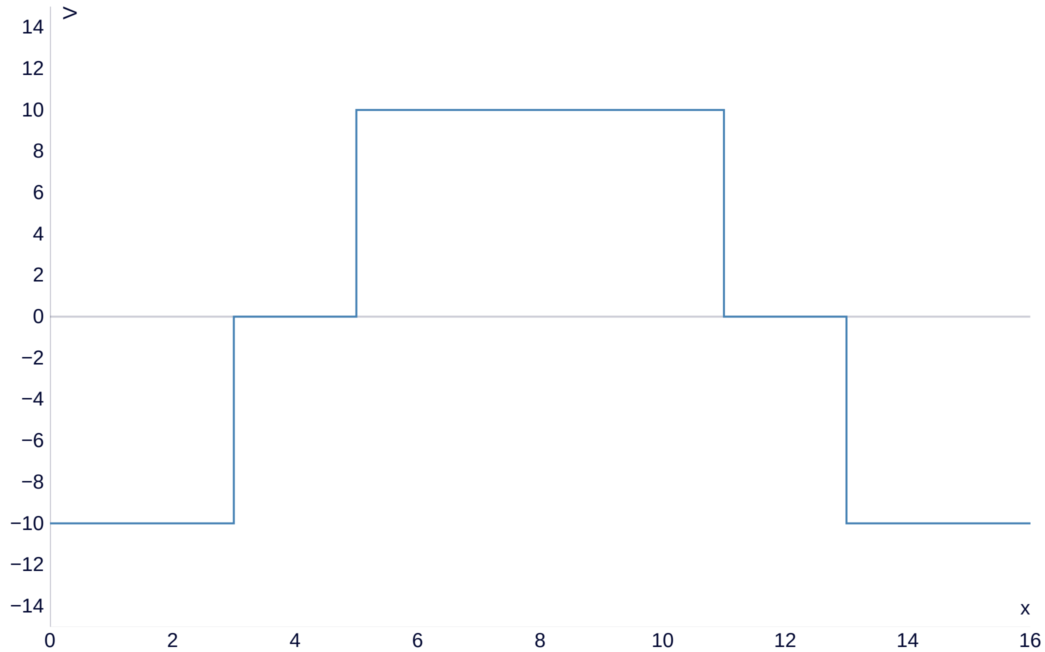
$$E_z = \frac{\rho_s h}{4\pi\epsilon_0} \int_{-d/2}^{d/2} \int_{-\infty}^{\infty} \frac{1}{(x^2+y^2+h^2)^{3/2}} \, dx \, dy$$
$$= \frac{\rho_s}{\pi\epsilon_0} \arctan\left(\frac{d}{2h}\right)$$

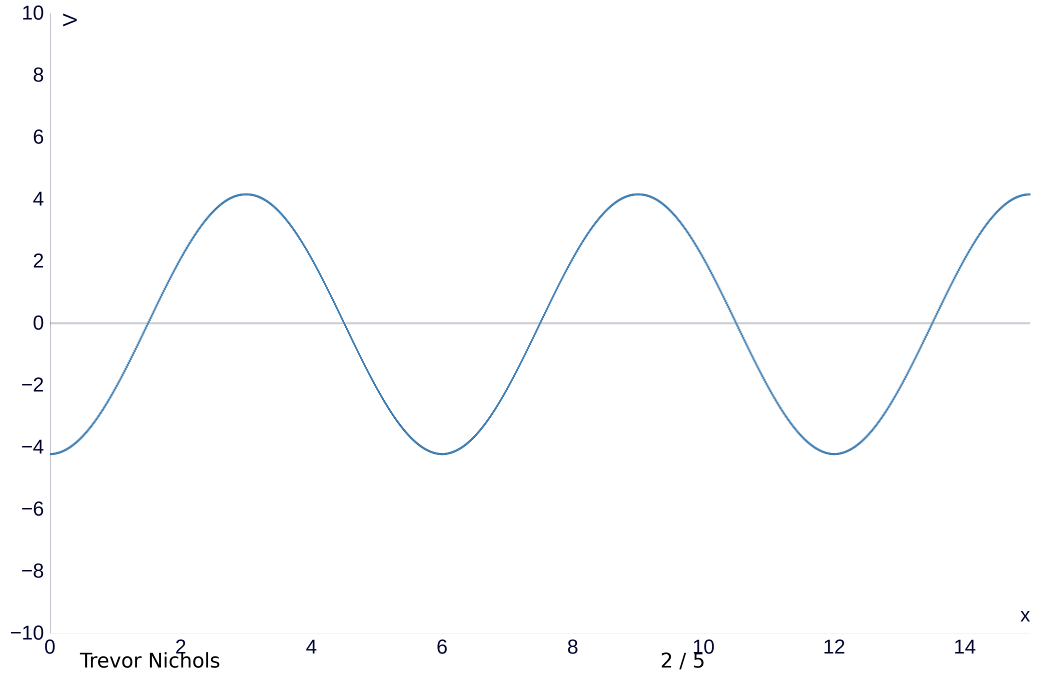
As $d \to \infty$

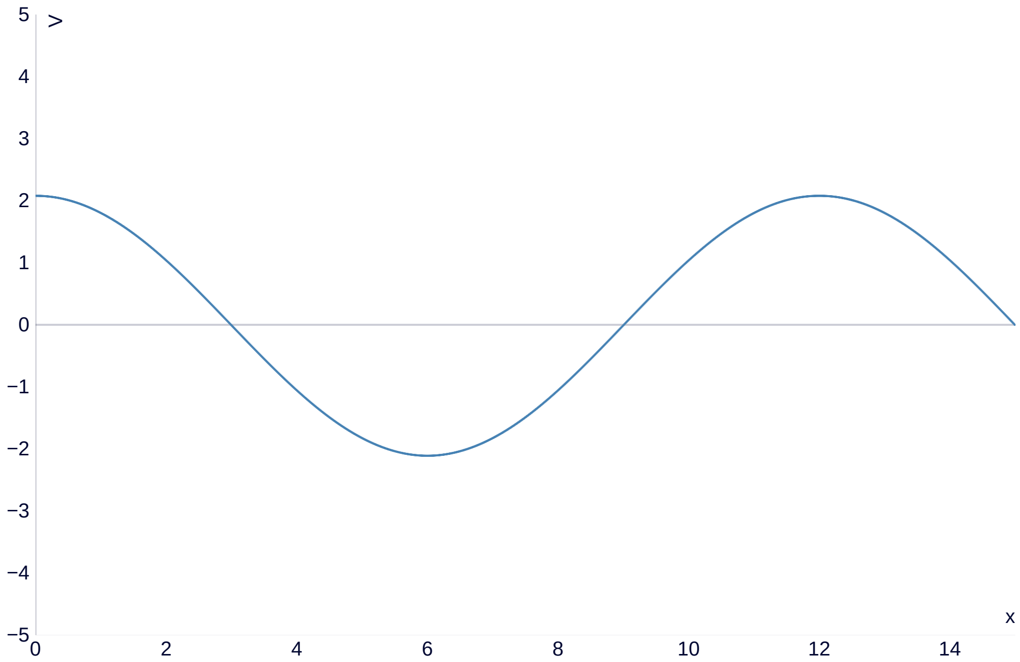
$$= \frac{\rho_s}{2\epsilon_0}$$

4

For each of the distributions of the elctric potential V show below, sketch the corresponding distrbution of E.







5

A cylindrical bar of silicon has a radius of 4mm and a length of 8cm. If a voltage of 5V is applied between the ends of the bar and $\mu_e = 0.13\ (m^2/Vs), \mu_h = 0.05\ (m^2/Vs)$, $N_e = 1.5 \times 10^{16}$ electrons/m3, and $N_h = N_e$, find the following:

a

The conductivity of silicon

Answer

$\sigma = qn(\mu_c + \mu_h) = 4.325 \times 10^{-4}\ S/m$

b

The current I flowing in the bar

Answer

$I = \sigma V \pi r^2 / L = 1.3587\ \mu A$

c

The drift velocities u_e and u_h

Answer

$u_e = \mu_e E = 8.125\ m/s$

$u_h = \mu_h E = 3.125\ m/s$

d

The resistance of the bar

> ✓ **Answer**
> $R = V/I$
> $= 3.68\ M\Omega$

e

The power dissipated in the bar

> ✓ **Answer**
> $P = VI$
> $= 6.79\ \mu W$

6

Determine the force of attraction in a parallel-plate capacitor with $A = 5\ cm^2$, $d = 2\ cm$, and $\epsilon_r = 4$ if the voltage across it is $50\ V$.

> ✓ **Answer**
> $F = \frac{\epsilon A V^2}{2d^2}$
> $F = 55.34\ nN$

7

The figure depicts a capacitor consisting of two parallel, conducting plates separated by a distance d . The space between the plates contains two adjacent dielectrics: one with permittivity ϵ_1 and surface area A_1 and another with ϵ_2 and A_2. The objective of the configuration shown in the figure is equivalent to two capacitances in parallel, as illustrated in the figure.

$C = C_1 + C_2$
$C_1 = \frac{\epsilon_1 A_1}{d}$
$C_2 = \frac{\epsilon_2 A_2}{d}$

a

Find the electric fields E_1 and E_2 in the two dielectric layers.

> ✓ **Answer**
> $E_1 = E_2 = V/d$ as the plates have equal potential across themselves.

b

Calculate the energy stored in each section and use the result to calculate C_1 and C_2.

✓ Answer

$U = \frac{\epsilon V^2}{2d^2} Ad = \frac{\epsilon V^2 A}{2d}$
$U = \frac{1}{2} CV^2$
$C = \frac{\epsilon A}{d}$
$C_1 = \frac{\epsilon_1 A_1}{d}$
$C_2 = \frac{\epsilon_2 A_2}{d}$

c

Use the total energy stored in the capacitor to obtain an expression for C. Show that the equation is a valid result.

✓ Answer

$U = \frac{\epsilon_1 V^2 A_1}{2d} + \frac{\epsilon_2 V^2 A_2}{2d}$
$= \frac{V^2}{2} (\frac{\epsilon_1 A_1}{d} + \frac{\epsilon_2 A_2}{d})$
$C = \frac{\epsilon_1 A_1}{d} + \frac{\epsilon_2 A_2}{d} = C_1 + C_2$